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Blockchain

A blockchain is a growing list of records, called blocks, that are linked together using cryptography. Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data (generally represented as a Merkle tree).
The timestamp proves that the transaction data existed when the block was published in order to get into its hash. As blocks each contain information about the block previous to it, they form a chain, with each additional block reinforcing the ones before it. Therefore, blockchains are resistant to modification of their data because once recorded, the data in any given block cannot be altered retroactively without altering all subsequent blocks.
Blockchains are typically managed by a peer-to-peer network for use as a publicly distributed ledger, where nodes collectively adhere to a protocol to communicate and validate new blocks. Although blockchain records are not unalterable as forks are possible, blockchains may be considered secure by design and exemplify a distributed computing system with high Byzantine fault tolerance.
The blockchain was invented by a person (or group of people) using the name Satoshi Nakamoto in 2008 to serve as the public transaction ledger of the cryptocurrency bitcoin. The identity of Satoshi Nakamoto remains unknown to date. The invention of the blockchain for bitcoin made it the first digital currency to solve the double-spending problem without the need of a trusted authority or central server. The bitcoin design has inspired other applications and blockchains that are readable by the public and are widely used by cryptocurrencies.
The blockchain is considered a type of payment rail.
Private blockchains have been proposed for business use but Computerworld called the marketing of such privatized blockchains without a proper security model "snake oil". However, others have argued that permissioned blockchains, if carefully designed, may be more decentralized and therefore more secure in practice than permissionless ones.

Blockchain History

Cryptographer David Chaum first proposed a blockchain-like protocol in his 1982 dissertation "Computer Systems Established, Maintained, and Trusted by Mutually Suspicious Groups." Further work on a cryptographically secured chain of blocks was described in 1991 by Stuart Haber and W. Scott Stornetta.They wanted to implement a system wherein document timestamps could not be tampered with. In 1992, Haber, Stornetta, and Dave Bayer incorporated Merkle trees to the design, which improved its efficiency by allowing several document certificates to be collected into one block.The first blockchain was conceptualized by a person (or group of people) known as Satoshi Nakamoto in 2008. Nakamoto improved the design in an important way using a Hashcash-like method to timestamp blocks without requiring them to be signed by a trusted party and introducing a difficulty parameter to stabilize rate with which blocks are added to the chain. The design was implemented the following year by Nakamoto as a core component of the cryptocurrency bitcoin, where it serves as the public ledger for all transactions on the network. In August 2014, the bitcoin blockchain file size, containing records of all transactions that have occurred on the network, reached 20 GB (gigabytes). In January 2015, the size had grown to almost 30 GB, and from January 2016 to January 2017, the bitcoin blockchain grew from 50 GB to 100 GB in size. The ledger size had exceeded 200 GiB by early 2020. The words block and chain were used separately in Satoshi Nakamoto's original paper, but were eventually popularized as a single word, blockchain, by 2016.According to Accenture, an application of the diffusion of innovations theory suggests that blockchains attained a 13.5% adoption rate within financial services in 2016, therefore reaching the early adopters phase. Industry trade groups joined to create the Global Blockchain Forum in 2016, an initiative of the Chamber of Digital Commerce. In May 2018, Gartner found that only 1% of CIOs indicated any kind of blockchain adoption within their organisations, and only 8% of CIOs were in the short-term "planning or [looking at] active experimentation with blockchain". For the year 2019 Gartner reported 5% of CIOs believed blockchain technology was a 'game-changer' for their business.

Structure

A blockchain is a decentralized, distributed, and oftentimes public, digital ledger consisting of records called blocks that is used to record transactions across many computers so that any involved block cannot be altered retroactively, without the alteration of all subsequent blocks. This allows the participants to verify and audit transactions independently and relatively inexpensively.A blockchain database is managed autonomously using a peer-to-peer network and a distributed timestamping server.They are authenticated by mass collaboration powered by collective self-interests. Such a design facilitates robust workflow where participants' uncertainty regarding data security is marginal. The use of a blockchain removes the characteristic of infinite reproducibility from a digital asset. It confirms that each unit of value was transferred only once, solving the long-standing problem of double spending. A blockchain has been described as a value-exchange protocol. A blockchain can maintain title rights because, when properly set up to detail the exchange agreement, it provides a record that compels offer and acceptance.Logically, a blockchain can be seen as consisting of several layers:

infrastructure (hardware)

networking (node discovery, information propagation and verification)

consensus (proof of work, proof of stake)

data (blocks, transactions)

application (smart contracts/decentralized applications, if applicable)

Blocks

Blocks hold batches of valid transactions that are hashed and encoded into a Merkle tree.Each block includes the cryptographic hash of the prior block in the blockchain, linking the two. The linked blocks form a chain.This iterative process confirms the integrity of the previous block, all the way back to the initial block, which is known as the genesis block.Sometimes separate blocks can be produced concurrently, creating a temporary fork. In addition to a secure hash-based history, any blockchain has a specified algorithm for scoring different versions of the history so that one with a higher score can be selected over others. Blocks not selected for inclusion in the chain are called orphan blocks.Peers supporting the database have different versions of the history from time to time. They keep only the highest-scoring version of the database known to them. Whenever a peer receives a higher-scoring version (usually the old version with a single new block added) they extend or overwrite their own database and retransmit the improvement to their peers. There is never an absolute guarantee that any particular entry will remain in the best version of the history forever.Blockchains are typically built to add the score of new blocks onto old blocks and are given incentives to extend with new blocks rather than overwrite old blocks. Therefore, the probability of an entry becoming superseded decreases exponentially as more blocks are built on top of it, eventually becoming very low.For example, bitcoin uses a proof-of-work system, where the chain with the most cumulative proof-of-work is considered the valid one by the network. There are a number of methods that can be used to demonstrate a sufficient level of computation. Within a blockchain the computation is carried out redundantly rather than in the traditional segregated and parallel manner.

Block time

The block time is the average time it takes for the network to generate one extra block in the blockchain. Some blockchains create a new block as frequently as every five seconds.By the time of block completion, the included data becomes verifiable.In cryptocurrency, this is practically when the transaction takes place, so a shorter block time means faster transactions. The block time for Ethereum is set to between 14 and 15 seconds, while for bitcoin it is on average 10 minutes.

Hard forks

A hard fork is a rule change such that the software validating according to the old rules will see the blocks produced according to the new rules as invalid. In case of a hard fork, all nodes meant to work in accordance with the new rules need to upgrade their software.If one group of nodes continues to use the old software while the other nodes use the new software, a permanent split can occur.For example, Ethereum has hard-forked to "make whole" the investors in The DAO, which had been hacked by exploiting a vulnerability in its code. In this case, the fork resulted in a split creating Ethereum and Ethereum Classic chains.
In 2014 the Nxt community was asked to consider a hard fork that would have led to a rollback of the blockchain records to mitigate the effects of a theft of 50 million NXT from a major cryptocurrency exchange.The hard fork proposal was rejected, and some of the funds were recovered after negotiations and ransom payment. Alternatively, to prevent a permanent split, a majority of nodes using the new software may return to the old rules, as was the case of bitcoin split on 12 March 2013. A more recent hard-fork example is of Bitcoin in 2017, which resulted in a split creating Bitcoin Cash. The network split was mainly due to a disagreement in how to increase the transactions per second to accommodate for demand.

Decentralization

By storing data across its peer-to-peer network, the blockchain eliminates a number of risks that come with data being held centrally.The decentralized blockchain may use ad hoc message passing and distributed networking. One risk of a lack of a decentralization is a so-called "51% attack" where a central entity can gain control of more than half of a network and can manipulate that specific blockchain record at-will, allowing double-spending.Peer-to-peer blockchain networks lack centralized points of vulnerability that computer crackers can exploit; likewise, it has no central point of failure. Blockchain security methods include the use of public-key cryptography. A public key (a long, random-looking string of numbers) is an address on the blockchain. Value tokens sent across the network are recorded as belonging to that address. A private key is like a password that gives its owner access to their digital assets or the means to otherwise interact with the various capabilities that blockchains now support. Data stored on the blockchain is generally considered incorruptible.Every node in a decentralized system has a copy of the blockchain. Data quality is maintained by massive database replication and computational trust. No centralized "official" copy exists and no user is "trusted" more than any other.Transactions are broadcast to the network using software. Messages are delivered on a best-effort basis. Mining nodes validate transactions,add them to the block they are building, and then broadcast the completed block to other nodes. Blockchains use various time-stamping schemes, such as proof-of-work, to serialize changes.Alternative consensus methods include proof-of-stake.Growth of a decentralized blockchain is accompanied by the risk of centralization because the computer resources required to process larger amounts of data become more expensive.

Openness

Open blockchains are more user-friendly than some traditional ownership records, which, while open to the public, still require physical access to view. Because all early blockchains were permissionless, controversy has arisen over the blockchain definition. An issue in this ongoing debate is whether a private system with verifiers tasked and authorized (permissioned) by a central authority should be considered a blockchain.Proponents of permissioned or private chains argue that the term "blockchain" may be applied to any data structure that batches data into time-stamped blocks. These blockchains serve as a distributed version of multiversion concurrency control (MVCC) in databases.Just as MVCC prevents two transactions from concurrently modifying a single object in a database, blockchains prevent two transactions from spending the same single output in a blockchain.Opponents say that permissioned systems resemble traditional corporate databases, not supporting decentralized data verification, and that such systems are not hardened against operator tampering and revision. Nikolai Hampton of Computerworld said that "many in-house blockchain solutions will be nothing more than cumbersome databases," and "without a clear security model, proprietary blockchains should be eyed with suspicion."

Permissionless

An advantage to an open, permissionless, or public, blockchain network is that guarding against bad actors is not required and no access control is needed.This means that applications can be added to the network without the approval or trust of others, using the blockchain as a transport layer.Bitcoin and other cryptocurrencies currently secure their blockchain by requiring new entries to include a proof of work. To prolong the blockchain, bitcoin uses Hashcash puzzles. While Hashcash was designed in 1997 by Adam Back, the original idea was first proposed by Cynthia Dwork and Moni Naor and Eli Ponyatovski in their 1992 paper "Pricing via Processing or Combatting Junk Mail".In 2016, venture capital investment for blockchain-related projects was weakening in the USA but increasing in China. Bitcoin and many other cryptocurrencies use open (public) blockchains. As of April 2018, bitcoin has the highest market capitalization.

Permissioned (private) blockchain

Permissioned blockchains use an access control layer to govern who has access to the network.In contrast to public blockchain networks, validators on private blockchain networks are vetted by the network owner. They do not rely on anonymous nodes to validate transactions nor do they benefit from the network effect.Permissioned blockchains can also go by the name of 'consortium' blockchains.It has been argued that permissioned blockchains can guarantee a certain level of decentralization, if carefully designed, as opposed to permissionless blockchains, which are often centralized in practice.

Disadvantages of private blockchain

Nikolai Hampton pointed out in Computerworld that "There is also no need for a '51 percent' attack on a private blockchain, as the private blockchain (most likely) already controls 100 percent of all block creation resources. If you could attack or damage the blockchain creation tools on a private corporate server, you could effectively control 100 percent of their network and alter transactions however you wished."This has a set of particularly profound adverse implications during a financial crisis or debt crisis like the financial crisis of 2007–08, where politically powerful actors may make decisions that favor some groups at the expense of others,and "the bitcoin blockchain is protected by the massive group mining effort. It's unlikely that any private blockchain will try to protect records using gigawatts of computing power — it's time consuming and expensive."He also said, "Within a private blockchain there is also no 'race'; there's no incentive to use more power or discover blocks faster than competitors. This means that many in-house blockchain solutions will be nothing more than cumbersome databases."



Blockchain analysis

The analysis of public blockchains has become increasingly important with the popularity of bitcoin, Ethereum, litecoin and other cryptocurrencies. A blockchain, if it is public, provides anyone who wants access to observe and analyse the chain data, given one has the know-how. The process of understanding and accessing the flow of crypto has been an issue for many cryptocurrencies, crypto-exchanges and banks. The reason for this is accusations of blockchain enabled cryptocurrencies enabling illicit dark market trade of drugs, weapons, money laundering etc. A common belief has been that cryptocurrency is private and untraceable, thus leading many actors to use it for illegal purposes. This is changing and now specialised tech-companies provide blockchain tracking services, making crypto exchanges, law-enforcement and banks more aware of what is happening with crypto funds and fiat crypto exchanges. The development, some argue, has led criminals to prioritise use of new cryptos such as Monero. The question is about public accessibility of blockchain data and the personal privacy of the very same data. It is a key debate in cryptocurrency and ultimately in blockchain.

Standardisation

There is a growing industrial need for blockchain standards because interoperability is considered critical to widespread adoption. Blockchain technologies show much potential as they provide capabilities that cannot normally be met in any other way if the requirement of interoperability between blockchains and with other technologies is met.In April 2016, Standards Australia submitted a proposal to the International Organization for Standardization to consider developing standards to support blockchain technology. This proposal resulted in the creation of ISO Technical Committee 307, Blockchain and Distributed Ledger Technologies. The technical committee has working groups relating to blockchain terminology, reference architecture, security and privacy, identity, smart contracts, governance and interoperability for blockchain and DLT, as well as standards specific to industry sectors and generic government requirements. More than 50 countries are participating in the standardization process together with external liaisons such as the Society for Worldwide Interbank Financial Telecommunication (SWIFT), the European Commission, the International Federation of Surveyors, the International Telecommunication Union (ITU) and the United Nations Economic Commission for Europe (UNECE). Many other national standards bodies and open standards bodies are also working on blockchain standards. These include the National Institute of Standards and Technology (NIST), the European Committee for Electrotechnical Standardization (CENELEC), the Institute of Electrical and Electronics Engineers (IEEE), the Organization for the Advancement of Structured Information Standards (OASIS), and the Internet Engineering Task Force (IETF).

Cryptocurrency

A cryptocurrency, crypto-currency, or crypto is a binary data designed to work as a medium of exchange wherein individual coin ownership records are stored in a ledger existing in a form of a computerized database using strong cryptography to secure transaction records, to control the creation of additional coins, and to verify the transfer of coin ownership. Some crypto schemes use validators to maintain the cryptocurrency. In a proof-of-stake model, owners put up their tokens as collateral. In return, they get authority over the token in proportion to the amount they stake. Generally, these token stakers get additional ownership in the token over time via network fees, newly minted tokens or other such reward mechanisms.Cryptocurrency does not exist in physical form (like paper money) and is typically not issued by a central authority. Cryptocurrencies typically use decentralized control as opposed to a central bank digital currency (CBDC).When a cryptocurrency is minted or created prior to issuance or issued by a single issuer, it is generally considered centralized. When implemented with decentralized control, each cryptocurrency works through distributed ledger technology, typically a blockchain, that serves as a public financial transaction database. Bitcoin, first released as open-source software in 2009, is the first decentralized cryptocurrency. Since the release of bitcoin, many other cryptocurrencies have been created.

History

In 1983, the American cryptographer David Chaum conceived an anonymous cryptographic electronic money called ecash. Later, in 1995, he implemented it through Digicash,an early form of cryptographic electronic payments which required user software in order to withdraw notes from a bank and designate specific encrypted keys before it can be sent to a recipient. This allowed the digital currency to be untraceable by the issuing bank, the government, or any third party. In 1996, the National Security Agency published a paper entitled How to Make a Mint: the Cryptography of Anonymous Electronic Cash, describing a Cryptocurrency system, first publishing it in an MIT mailing list and later in 1997, in The American Law Review (Vol. 46, Issue 4). In 1998, Wei Dai published a description of "b-money", characterized as an anonymous, distributed electronic cash system. Shortly thereafter, Nick Szabo described bit gold.Like bitcoin and other cryptocurrencies that would follow it, bit gold (not to be confused with the later gold-based exchange, BitGold) was described as an electronic currency system which required users to complete a proof of work function with solutions being cryptographically put together and published. In 2009, the first decentralized cryptocurrency, bitcoin, was created by presumably pseudonymous developer Satoshi Nakamoto. It used SHA-256, a cryptographic hash function, in its proof-of-work scheme. In April 2011, Namecoin was created as an attempt at forming a decentralized DNS, which would make internet censorship very difficult. Soon after, in October 2011, Litecoin was released. It used scrypt as its hash function instead of SHA-256. Another notable cryptocurrency, Peercoin, used a proof-of-work/proof-of-stake hybrid. On 6 August 2014, the UK announced its Treasury had commissioned a study of cryptocurrencies, and what role, if any, they could play in the UK economy. The study was also to report on whether regulation should be considered. In June 2021, El Salvador became the first country to accept Bitcoin as legal tender, after the Legislative Assembly had voted 62–22 to pass a bill submitted by President Nayib Bukele classifying the cryptocurrency as such.

Altcoins

Tokens, cryptocurrencies, and other types of digital assets that are not bitcoin are collectively known as alternative cryptocurrencies,typically shortened to "altcoins" or "alt coins".Paul Vigna of The Wall Street Journal also described altcoins as "alternative versions of bitcoin"given its role as the model protocol for altcoin designers. The term is commonly used to describe coins and tokens created after bitcoin. A list of some cryptocurrencies can be found in the List of cryptocurrencies article. Altcoins often have underlying differences with bitcoin. For example, Litecoin aims to process a block every 2.5 minutes, rather than bitcoin's 10 minutes, which allows Litecoin to confirm transactions faster than bitcoin. Another example is Ethereum, which has smart contract functionality that allows decentralized applications to be run on its blockchain.Ethereum was the most used blockchain in 2020, according to Bloomberg News. In 2016, it had the largest "following" of any altcoin, according to the New York Times. Significant rallies across altcoin markets are often referred to as an "altseason".

Crypto token

A blockchain account can provide functions other than making payments, for example in decentralized applications or smart contracts. (Units of) fungible tokens are sometimes referred to as crypto tokens (or cryptotokens). These terms are usually reserved for other fungible tokens than the main cryptocurrency of the blockchain, that is, usually, for fungible tokens issued within a smart contract running on top of a blockchain such as Ethereum.There are also non-fungible tokens. Tokens are risks. You cannot blame the token owners when there is a problem (including the Archive). Invest by accepting the risk.

Architecture

Decentralized cryptocurrency is produced by the entire cryptocurrency system collectively, at a rate which is defined when the system is created and which is publicly known. In centralized banking and economic systems such as the Federal Reserve System, corporate boards or governments control the supply of currency by printing units of fiat money or demanding additions to digital banking ledgers. In the case of decentralized cryptocurrency, companies or governments cannot produce new units, and have not so far provided backing for other firms, banks or corporate entities which hold asset value measured in it. The underlying technical system upon which decentralized cryptocurrencies are based was created by the group or individual known as Satoshi Nakamoto. As of May 2018, over 1,800 cryptocurrency specifications existed. Within a proof-of-work cryptocurrency system such as Bitcoin, the safety, integrity and balance of ledgers is maintained by a community of mutually distrustful parties referred to as miners: who use their computers to help validate and timestamp transactions, adding them to the ledger in accordance with a particular timestamping scheme. In a proof-of-stake (PoS) blockchain, transactions are validated by holders of the associated cryptocurrency, sometimes grouped together in stake pools. Most cryptocurrencies are designed to gradually decrease the production of that currency, placing a cap on the total amount of that currency that will ever be in circulation. Compared with ordinary currencies held by financial institutions or kept as cash on hand, cryptocurrencies can be more difficult for seizure by law enforcement.

Nodes

In the world of Cryptocurrency, a node is a computer that connects to a cryptocurrency network. The node supports the relevant cryptocurrency's network through either; relaying transactions, validation or hosting a copy of the blockchain. In terms of relaying transactions each network computer (node) has a copy of the blockchain of the cryptocurrency it supports, when a transaction is made the node creating the transaction broadcasts details of the transaction using encryption to other nodes throughout the node network so that the transaction (and every other transaction) is known. Node owners are either volunteers, those hosted by the organisation or body responsible for developing the cryptocurrency blockchain network technology or those that are enticed to host a node to receive rewards from hosting the node network.

Timestamping

Cryptocurrencies use various timestamping schemes to "prove" the validity of transactions added to the blockchain ledger without the need for a trusted third party. The first timestamping scheme invented was the proof-of-work scheme. The most widely used proof-of-work schemes are based on SHA-256 and scrypt.Some other hashing algorithms that are used for proof-of-work include CryptoNight, Blake, SHA-3, and X11. The proof-of-stake is a method of securing a cryptocurrency network and achieving distributed consensus through requesting users to show ownership of a certain amount of currency. It is different from proof-of-work systems that run difficult hashing algorithms to validate electronic transactions. The scheme is largely dependent on the coin, and there's currently no standard form of it. Some cryptocurrencies use a combined proof-of-work and proof-of-stake scheme.

Mining

In cryptocurrency networks, mining is a validation of transactions. For this effort, successful miners obtain new cryptocurrency as a reward. The reward decreases transaction fees by creating a complementary incentive to contribute to the processing power of the network. The rate of generating hashes, which validate any transaction, has been increased by the use of specialized machines such as FPGAs and ASICs running complex hashing algorithms like SHA-256 and scrypt. This arms race for cheaper-yet-efficient machines has existed since the day the first cryptocurrency, bitcoin, was introduced in 2009. With more people venturing into the world of virtual currency, generating hashes for this validation has become far more complex over the years, with miners having to invest large sums of money on employing multiple high performance ASICs. Thus the value of the currency obtained for finding a hash often does not justify the amount of money spent on setting up the machines, the cooling facilities to overcome the heat they produce, and the electricity required to run them. Favorite regions for mining are those with cheap electricity or a cold climate. As of July 2019, bitcoin's electricity consumption is estimated to about 7 gigawatts, 0.2% of the global total, or equivalent to that of Switzerland. Some miners pool resources, sharing their processing power over a network to split the reward equally, according to the amount of work they contributed to the probability of finding a block. A "share" is awarded to members of the mining pool who present a valid partial proof-of-work. As of February 2018, the Chinese Government halted trading of virtual currency, banned initial coin offerings and shut down mining. Some Chinese miners have since relocated to Canada. One company is operating data centers for mining operations at Canadian oil and gas field sites, due to low gas prices. In June 2018, Hydro Quebec proposed to the provincial government to allocate 500 MW to crypto companies for mining. According to a February 2018 report from Fortune, Iceland has become a haven for cryptocurrency miners in part because of its cheap electricity. In March 2018, the city of Plattsburgh in upstate New York put an 18-month moratorium on all cryptocurrency mining in an effort to preserve natural resources and the "character and direction" of the city.

Wallets

A cryptocurrency wallet stores the public and private "keys" (address) or seed which can be used to receive or spend the cryptocurrency. With the private key, it is possible to write in the public ledger, effectively spending the associated cryptocurrency. With the public key, it is possible for others to send currency to the wallet. There exist multiple methods of storing keys or seed in a wallet from using paper wallets which are traditional public, private or seed keys written on paper to using hardware wallets which are dedicated hardware to securely store your wallet information, using a digital wallet which is a computer with a software hosting your wallet information, hosting your wallet using an exchange where cryptocurrency is traded. or by storing your wallet information on a digital medium such as plaintext.

Anonymity

Bitcoin is pseudonymous rather than anonymous in that the cryptocurrency within a wallet is not tied to people, but rather to one or more specific keys (or "addresses"). Thereby, bitcoin owners are not identifiable, but all transactions are publicly available in the blockchain. Still, cryptocurrency exchanges are often required by law to collect the personal information of their users. Additions such as Monero, Zerocoin, Zerocash and CryptoNote have been suggested, which would allow for additional anonymity and fungibility.

Block rewards

Proof-of-work cryptocurrencies, such as bitcoin, offer block rewards incentives for miners. There has been an implicit belief that whether miners are paid by block rewards or transaction fees does not affect the security of the blockchain, but a study suggests that this may not be the case under certain circumstances. The rewards paid to miners increase the supply of the cryptocurrency. By making sure that verifying transactions is a costly business, the integrity of the network can be preserved as long as benevolent nodes control a majority of computing power. The verification algorithm requires a lot of processing power, and thus electricity in order to make verification costly enough to accurately validate public blockchain. Not only do miners have to factor in the costs associated with expensive equipment necessary to stand a chance of solving a hash problem, they further must consider the significant amount of electrical power in search of the solution. Generally, the block rewards outweigh electricity and equipment costs, but this may not always be the case. The current value, not the long-term value, of the cryptocurrency supports the reward scheme to incentivize miners to engage in costly mining activities. Some sources claim that the current bitcoin design is very inefficient, generating a welfare loss of 1.4% relative to an efficient cash system. The main source for this inefficiency is the large mining cost, which is estimated to be US$360 Million per year. This translates into users being willing to accept a cash system with an inflation rate of 230% before being better off using bitcoin as a means of payment. However, the efficiency of the bitcoin system can be significantly improved by optimizing the rate of coin creation and minimizing transaction fees. Another potential improvement is to eliminate inefficient mining activities by changing the consensus protocol altogether.

Initial coin offerings

Transaction fees for cryptocurrency depend mainly on the supply of network capacity at the time, versus the demand from the currency holder for a faster transaction. The currency holder can choose a specific transaction fee, while network entities process transactions in order of highest offered fee to lowest. Cryptocurrency exchanges can simplify the process for currency holders by offering priority alternatives and thereby determine which fee will likely cause the transaction to be processed in the requested time. For ether, transaction fees differ by computational complexity, bandwidth use, and storage needs, while bitcoin transaction fees differ by transaction size and whether the transaction uses SegWit. In September 2018, the median transaction fee for ether corresponded to $0.017,while for bitcoin it corresponded to $0.55. Some cryptocurrencies have no transaction fees, and instead rely on client-side proof-of-work as the transaction prioritization and anti-spam mechanism.

Exchanges

Cryptocurrency exchanges allow customers to trade cryptocurrencies for other assets, such as conventional fiat money, or to trade between different digital currencies.

Atomic swaps

Atomic swaps are a mechanism where one cryptocurrency can be exchanged directly for another cryptocurrency, without the need for a trusted third party such as an exchange.

ATMs

Jordan Kelley, founder of Robocoin, launched the first bitcoin ATM in the United States on 20 February 2014. The kiosk installed in Austin, Texas, is similar to bank ATMs but has scanners to read government-issued identification such as a driver's license or a passport to confirm users' identities.

Initial coin offerings

An initial coin offering (ICO) is a controversial means of raising funds for a new cryptocurrency venture. An ICO may be used by startups with the intention of avoiding regulation. However, securities regulators in many jurisdictions, including in the U.S., and Canada, have indicated that if a coin or token is an "investment contract" (e.g., under the Howey test, i.e., an investment of money with a reasonable expectation of profit based significantly on the entrepreneurial or managerial efforts of others), it is a security and is subject to securities regulation. In an ICO campaign, a percentage of the cryptocurrency (usually in the form of "tokens") is sold to early backers of the project in exchange for legal tender or other cryptocurrencies, often bitcoin or ether. According to PricewaterhouseCoopers, four of the 10 biggest proposed initial coin offerings have used Switzerland as a base, where they are frequently registered as non-profit foundations. The Swiss regulatory agency FINMA stated that it would take a "balanced approach" to ICO projects and would allow "legitimate innovators to navigate the regulatory landscape and so launch their projects in a way consistent with national laws protecting investors and the integrity of the financial system." In response to numerous requests by industry representatives, a legislative ICO working group began to issue legal guidelines in 2018, which are intended to remove uncertainty from cryptocurrency offerings and to establish sustainable business practices. The twenty-page section up to this point is an excerpt from Wikipedia. We wanted to give accurate information about these issues.

Archive Token

It was created as a payment tool for archive cloud systems. There are a total of one hundred million(100.000.000) archival tokens. Archive token mining is not allowed. The abbreviation for archive token is "ARH". Archive is binance smart chain token. The aim of our project is to combine blockchain, which is the technology of the future, with currently used cloud systems. We believe that our archive token, which was issued for the purpose of paying storage fees, will have great potential both now and for the future, and we aim to do so. Archive token, we will try to make it the best. For this, we will constantly renew our team and make it the best (except the boss). Our project will work to achieve one or more of the following objectives, provided the necessary conditions are met.

-Storing information via our own website and receiving payments with archive tokens

-Making the association with a cloud service company and completing the archive token on the payment path

-Buying a cloud service company and getting paid with archive token only

-R & D work, by making the necessary team upgrade, to ensure that the information of the people is stored with blockchain technology and to receive the payment only with archive tokens.

-To ensure that the information of corporate companies is stored with blockchain technology and to be paid only with archive tokens.

Achieving aims will take time. Just invest in advance and wait patiently.